GM ASEP / ASE A2 / VERSION 2.0

A2 SIMPLE SHEET

The middle layer: enough reasoning for a Tech A/B or most-likely stem, weighted to the official areas. [G x.x] = study guide section.

Prepared October 14, 2026 **Exam** October 15, 2026

Scope ASE A2, official 2026 task list (45 scored questions)

CONTENTS

00 How to Use This Sheet

01 Area A: Diagnosis (58%, 26 questions)

02 Area B: In-Vehicle Maintenance and Repair (27%, 12 questions)

03 Area C: Removal, Inspection, Installation (15%, 7 questions)

04 Your Weak Spots (Unit 1 quiz and Unit 2 test)

05 Wrong-Direction Traps

00 HOW TO USE THIS SHEET

This is the middle layer between the 34-page study guide and the cram sheet. It has enough explanation to reason through a Tech A/B or "most likely" stem, without the guide's depth. Everything here comes from the study guide; section numbers in brackets (**[G 2.4]**) point to the full explanation.

The exam: 55 questions, **45 scored**, 75 minutes, about 80 seconds each. Unscored items aren't marked, so treat every one as scored. No penalty for guessing: never leave one blank.

AREA	SCORED QS	SHARE	QUESTIONS PER TASK	WHAT IT REWARDS
A. Diagnosis	26	**58%**	~0.9	Symptom → cause → the check that separates causes
B. In-vehicle maintenance and repair	12	**27%**	**1.0 (best yield)**	Verify the cheap root cause before replacing the expensive part
C. Removal, inspection, installation	7	15%	~0.3	Order of steps and classic mistakes, not specs

What's left:

WHEN	DO THIS
Wed night (after the class final)	Read this sheet start to finish. Extra time on **A.3 (apply chart + planetary)** and **A.5 (stall)**, your two confirmed miss areas. Open the guide section in brackets for anything that doesn't click. Then the trap list in Section 5

WHEN	DO THIS
Thu morning	Cram sheet only. Stop early

---

01 AREA A: DIAGNOSIS (58%, 26 QUESTIONS)

A.1 THE METHOD THAT ANSWERS "FIRST" AND "NEXT" STEMS

Diagnostic order [G 7.4]: verify the concern → fluid level, condition and type (the SI way) → SI, bulletins, DTCs (the code is the starting point, not the diagnosis) → rule out engine, inputs, cooler and electrical supply → road test and scan data → **line pressure** → stall test only if the OEM publishes one → repair, clear with the scan tool, relearn, **verify**.

GM strategy-based diagnosis [G 8]: 1 verify the concern → 2 is it operating as designed? (if **yes, stop and explain**, compare with an identical vehicle) → 3 preliminary checks (visual, history, sounds and smells, DTCs) → 4 Diagnostic System Check → 5 bulletins, recalls, **software** (check for an update **before** replacing a module) → 6 diagnostic path: **6.1 current DTC**, **6.2 symptom no DTC**, **6.3 no published diagnostic** (build a plan from schematics), **6.4 intermittent/history** → 7 isolate, repair, **verify** → 8 re-examine; not fixed = **back to step 1**. A "what next" stem is asking where you are on this chart.

Question words: *could cause* = one valid mechanism is enough. *Most likely* = the choice that explains **every** symptom in the stem. *First/next* = cheapest, least invasive, most informative. *EXCEPT* = mark each "yes/no"; the single "no" wins. *Tech A/B* = judge each alone; "both" and "neither" are real answers.

Book answer beats shop habit. Clear codes with the scan tool (never a battery pull). Check the level the SI way. Test before replacing. Reset adapts is not a repair.

A.2 CLASSIFY THE SYMPTOM FIRST [G 2]

Before reading the choices, ask what the symptom **follows**:

FOLLOWS…	THINK FIRST OF…
Engine rpm (also in P/N)	Pump, converter, flexplate
Road speed	Gear train, output, final drive, tires, driveline
One gear or range only	The element **unique** to that range (apply chart)
Every range	Fluid, pump/pressure, input clutch, manual valve/linkage
Cold only, clears warm	Fluid level, thick-fluid or bore-leak effects
First engagement after sitting	Converter drain-back
Engine data or wheel-speed data	**Not the transmission** (misfire, TPS, MAF, ABS, tire size, mounts)

A.3 APPLY CHART AND PLANETARY: YOUR BIGGEST RISK [G 1]

The names (Unit 1 misses): sun = small center gear, external teeth. **Carrier holds the planets** (it is not the big outer gear). **Ring = big outer gear with internal teeth = the "internal gear."** The sun is **never** the internal gear.

The rules: one member **driven**, one **held**, one **output**. Find the carrier: **output = reduction, input = overdrive, held = reverse.** Any **two** members driven together = **direct 1:1**. Nothing held or a member disconnected = **neutral**.

INPUT	HELD	OUTPUT	RESULT	30/70 EXAMPLE
Sun	Ring	Carrier	Reduction (steepest)	3.33:1
Ring	Sun	Carrier	Reduction	1.43:1
Carrier	Sun	Ring	Overdrive	0.70:1
Carrier	Ring	Sun	Overdrive (steepest)	0.30:1
Sun	Carrier	Ring	**Reverse**, reduction	2.33:1
Ring	Carrier	Sun	**Reverse**, overdrive	0.43:1
Any two driven	Nothing	Whole set	Direct	1:1

Swap input and output and the ratio flips to its reciprocal. **Ratios in series multiply, never add** (2:1 then 3:1 = 6:1; a big number is not a mistake). **Ravigneaux** = two suns, two planet sets, one carrier, one ring. **LePelletier** = simple planetary + Ravigneaux. **Reaction (holding) clutch:** steel plates splined to the **case**.

Apply-chart elimination [G 1.3]:

1. Sort ranges into **affected** and **working** (include Park, Neutral, manual ranges, engine braking).
2. Several failed ranges → the element applied in **all** of them. One failed range → the element used **only** there.
3. Anything applied in a **working** range is cleared.
4. **Slip, flare, no drive** = an element that fails to **hold**. **Tie-up, bind, braked feel** = an element **seized or not releasing**, showing up where the chart says it should be off.
5. **Slip in D1 but fine in manual 1** = the one-way clutch (manual 1 backs it up with a friction element).

Worked example: no 3rd and no reverse, 1st and 2nd fine. 3rd = forward + direct; reverse = direct + low/reverse. Common to both: **direct**. Forward is cleared because it works in 1st and 2nd.

> **ASE TRAP** a first-gear-only one-way clutch can't cause slip in **every** forward gear. Slip in all forward gears with reverse OK = forward/input clutch or its circuit. A seized element binds; it doesn't leave you without a range.

A.4 LINE PRESSURE [G 2.4]

Order: fluid → line pressure → stall only if SI has one. **Low line pressure: don't stall test.**

READING	MOST LIKELY
Low in all ranges	Common supply: low fluid, clogged filter or suction air leak, worn pump, weak/stuck regulator, big common leak
Low in one range	Leak in the circuit fed only in that range (piston seal, sealing ring, servo seal)
High in all ranges	Stuck regulator; **EPC in failsafe** (unplugged/open = **maximum**); TV cable tight or modulator vacuum lost (older units)
Normal pressure, no engagement	Routing (manual valve, shift valve, check ball) or mechanical (broken clutch, sprag)

- **Reverse reads higher by design.** Low in reverse only = reverse circuit.

- **Classic EPC: more current = lower pressure.** The scan tool shows the **command**; only a gauge shows **pressure**.
- **Vacuum modulator:** low vacuum (load, leak, disconnected hose) = high pressure, firm late shifts. Ruptured diaphragm = ATF in the vacuum hose, white smoke, fluid loss with no visible leak.

A.5 TORQUE CONVERTER, STALL AND TCC [G 2.6, 3.5]

Impeller = engine. Turbine = input shaft. Engine rpm = impeller; ISS = turbine. **TCC slip = engine rpm − turbine rpm.** Most heat comes from the converter, worst at stall.

	STATOR FREEWHEELS (WON'T HOLD)	STATOR SEIZED (WON'T RELEASE)
Launch	**Poor**	Normal
Highway	Normal (above ~35 mph)	**Poor**, overheats
Stall test	**Low**	Not diagnostic

> LEARNER (MISSED BOTH ON THE UNIT 2 TEST) **HIGH stall = transmission only. LOW stall = engine OR stator.** High stall, "stator" vs. "slipping clutch": **B only**. Low stall, "stator" vs. "engine": **both**.

- **High stall = a transmission element slipping.** Nothing inside the converter raises stall. "200 rpm high = stator" is backwards.
- **Low stall** = weak engine (rule out first, incl. restricted exhaust) or freewheeling stator.
- **TCC stuck on:** engine stalls or nearly stalls at a stop; slip ~0 at a stop.
- **TCC commanded on, slip stays high** = the **apply side** isn't holding. Check apply pressure and leaks before the piston (TCC valve bores, stator-support bushing, turbine-shaft seal) before condemning the converter.
- **TCC won't apply** usually means an **input** blocks it: brake switch reads "applied," TP dead spot, cold-reading temp sensor, no speed signal.
- **Shudder vs. misfire:** true TCC shudder is gone with the TCC **off** and gone fully **locked**. A misfire follows firing frequency, shows in misfire data, and is often **worse** locked. **Fluid** is the first suspect for shudder.

A.6 SHIFT QUALITY [G 2.2]

Every shift is a handoff.

TERM	CAUSE	RPM
Flare	Off-going releases **before** on-coming holds	Rises briefly
Tie-up = bind-up	Overlap: two ratios at once, or an element won't release	Pulled **down**, output braked
Slide-bump	Accumulator bottoms out before the shift ends	Smooth start, harsh finish
One harsh shift	That shift's accumulator (spring, piston, seal)	—
All shifts harsh	High line pressure	—

Noise: whine in P/N that rises with rpm = pump (or clogged filter, air at pickup). Clunk only after an auto-restart = stop/start hold system.

A.7 ELECTRONIC: COMMANDED VS. ACTUAL [G 3]

Your strongest area. One habit answers most items: **what did the TCM command, and what actually happened?**

AFTER THE COMMAND	AREA
Current tracks command, pressure/ratio responds	Output path fine: look at inputs, adapts, intermittents
Current tracks command, no pressure/ratio change	**Hydraulic or mechanical**
Current doesn't track, circuit DTC	**Electrical** (coil, harness, connector, driver)
No command sent	Enable conditions or inputs (TFT, OSS, TPS, brake switch, range)

> **ASE TRAP (YOUR RISK)** "circuit tests good, so the solenoid is good." A **performance** DTC (ratio, slip, stuck on/off) with a good circuit = **hydraulic or mechanical**. A solenoid click only proves the coil moved.

- **TFT (NTC thermistor): open = reads very cold** (P0713, high signal voltage). **Short to ground = reads very hot** (P0712).
- **ISS/OSS:** Hall-effect: **no ohmmeter**; check reference, ground, and scope the square wave. Ratio = ISS ÷ OSS. A bad signal gives ratio DTCs on a healthy unit.
- **TR/MLP:** compare range PID with the physical lever. Bad = no-crank in Park, wrong display, default gear.
- **Voltage drop under load,** not ohms. **Several U-codes at once** = battery, power, ground or bus first.
- **OBD II:** MIL off after 3 clean trips; history erases after 40 warm-ups (OEMs often say 80 for misfire/fuel). **Permanent DTC: not cleared by scan tool or battery pull**; it clears when the monitor passes. **P0700** = go scan the TCM.
- **Adapts:** clearing is **not** a repair. **Relearn** after TCM, valve body or transmission replacement; load solenoid characterization data. Programming needs a **battery maintainer, not a charger**.

A.8 NOT THE TRANSMISSION [G 2.7, 3.10]

Misfire mimics TCC shudder. TPS reading closed = early soft shifts; open = late harsh. Wheel-speed/ABS failure = no upshift, dead speedometer, no TCC. Traction control feels like slip but the ratio stays correct. Mismatched tires bind AWD. Rough shifts right after a reflash = zeroed adapts. **The answer must explain every symptom in the stem.**

A.9 OTHER TYPES: RECOGNIZE ONLY [G 6]

DCT: no converter, no stall test, no TCC; odd/even shaft split is its "apply chart"; judder from a stop = launch clutch; dry clutch has **no fluid to change**. **CVT:** pulleys and belt/chain; never generic ATF; one-direction ratio fault = that direction's primary circuit; low clamp = belt slip. **Don't carry one type's logic into another.**

---

02 AREA B: IN-VEHICLE MAINTENANCE AND REPAIR (27%, 12 QUESTIONS)

Best yield on the test: one question per task. The pattern everywhere: **verify the cheap root cause before replacing the expensive part.**

SHOP SHORTCUT	ASE WANTS
Replace the TR switch	Check cable/linkage and detent first, then index the switch
Replace the tailshaft seal	Check the **bushing** and yoke seal surface too

SHOP SHORTCUT	ASE WANTS
Flush milky fluid	Replace the cooler, rebuild (seals, frictions, converter), then flush
Replace the TCM with ATF in the connector	Fix the leaking case pass-through too
Add fluid because it slips	Check the level the SI way: overfull slips too

Fluid level, no dipstick [G 4.8]: scan-tool fluid temperature **inside the SI window**, **vehicle level**, engine idling, shift through the ranges, open the check plug. **Drip = correct, stream = overfull, nothing = low.** Topped off cold = overfull hot (ATF expands). **Overfull:** foam, slip, heat, fluid out the vent. **Underfull:** pump pulls air, delayed engagement, whine, slip worse cold or in turns. **Pan drop** replaces only part of the fluid.

Fluid condition [G 2.1]: dark/burnt = overheated or burned frictions. **Milky pink = coolant from the radiator cooler** (cooling system pressure pushes it through a tiny leak). Even a little damages frictions.

Pan debris [G 4.3]: a little gray paste on the magnet = normal. Steel = magnetic (gears, bearings). **Brass and aluminum don't stick to the magnet**: a clean magnet doesn't mean no metal.

Cooling [G 4.4]: overheating with **normal shifting** = restricted cooler/lines or **TBV stuck in bypass** (works like a thermostat). After any failure, flush and **flow-test warm with the transmission's own pump**.

Valve body [G 4.5]: separator plates and gaskets are application-specific (compare hole for hole). Missing check balls = no shift or no move. Torque **center outward** with an inch-pound wrench. **A new problem right after valve body work = the job itself** (torque, check balls, gasket), not a new failure.

Linkage and TR switch [G 4.1]: linkage → switch → module. Cranks in N but not P = switch index or cable. Creep in neutral or delayed engagement = manual valve between detents.

Leaks [G 4.2]: clean, dye, find the **highest** wet point. Bellhousing = front seal (or engine rear main; identify the fluid). Rear = tailshaft seal **and** bushing/yoke.

Electrical [G 4.6]: ATF wicks up the harness from a failed pass-through. Fretting corrosion causes intermittents. Don't spread terminals with meter probes. **Mounts [G 4.7]:** clunk on engagement, vibration; settle the powertrain before torquing.

---

03 AREA C: REMOVAL, INSPECTION, INSTALLATION (15%, 7 QUESTIONS)

Only ~0.3 questions per task. Learn the **order and the classic mistakes**; numbers are SI.

- **Converter install [G 5.1]:** turn it while pushing in until it drops through **stages** (input splines, stator splines, pump lugs). Free play to the flexplate before bolting. **Never pull the unit in with the bellhousing bolts.** Dowels present. **Front pump damage right after an R&I = installation first.** Flexplate cracks knock like an engine; don't impact the bolts. Cracked flexplate or case = **alignment** (reinstall the dowels).
- **After install [G 5.2]:** flush and flow-check the cooler, load calibration/characterization data, **relearn**, check level at SI temperature.
- **Clutch packs [G 5.4]:** soak new frictions. Lip seals face **toward the pressure**. Piston check ball stuck = burned clutch.
- **Air test** with low, clean, dry air: **thud = good, hiss = leak.**
- **One-way clutches can go in backward:** no drive in the range they carry, or a tie-up. Bench-check rotation.
- **Thrust washers:** petroleum jelly, **not white grease**. **End play:** dial indicator, correct with selective washers or snap rings. **Pump clearance:** straightedge and feeler gauge.
- **Bands:** a worn drum lets the band run out of travel: replace **both**.

---

04 YOUR WEAK SPOTS (UNIT 1 QUIZ AND UNIT 2 TEST)

MISSED	CORRECT
Carrier vs. ring layout (twice)	Carrier **holds the planets**; ring is the big outer gear
"Sun gear = internal gear"	**False.** Ring = internal gear
Ravigneaux	2 suns, 2 planet sets, 1 carrier, 1 ring
Total ratio	**Multiply** stages
Reaction clutch steel plates	Splined to the **case**
Disconnected member	**Neutral**, no torque flow
Stall HIGH, Tech A stator / Tech B slipping clutch (Unit 2: chose both)	**B only.** Converter can't raise stall
Stall LOW, Tech A stator / Tech B engine (Unit 2: chose neither)	**Both.** Weak engine or stator not holding

05 WRONG-DIRECTION TRAPS

Each is a place where a flashcard, website or shop slang has it backwards.

1. **TFT:** open = reads **cold**; short to ground = reads **hot**.
2. **Stall:** high = **transmission only**; low = engine **or** stator (your Unit 2 miss).
3. **Stator:** freewheeling = poor launch; seized = poor top end + heat.
4. **Order:** fluid → line pressure → stall. Never stall-test with low line pressure.
5. **Codes:** clear with the scan tool; a battery pull doesn't clear a permanent DTC.
6. **Adapts:** reset is verification, not repair.
7. **EPC:** more current = **less** pressure; unplugged = **maximum** (classic design; read the stem).
8. **Flare vs. tie-up:** opposites. Flare rpm up; tie-up output braked.
9. **First-gear one-way clutch** can't cause slip in every forward gear.
10. **Milky ATF:** replace cooler + rebuild + flush. "Flush and refill" is the distractor.
11. **Pan drop** changes only part of the fluid.
12. **Ratios in series multiply.**
13. **Circuit good ≠ solenoid good** when the DTC is a performance code.
14. **Operating as designed** ends the diagnosis at strategy step 2: explain it, don't replace parts.

Private study material. Not ASE content.